

MINISTRY OF EDUCATION AND TRAINING
EASTERN INTERNATIONAL UNIVERSITY



BUS 443

PRINCIPLES OF DATA ANALYTICS

Group Project

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1. Introduction

1.1 Business Background

Supply chain management plays a critical role in modern business operations by coordinating the flow of products, information, and financial resources from suppliers to end customers. As globalization and e-commerce continue to expand, organizations generate enormous volumes of transactional data related to customer orders, product sales, inventory, shipping, and logistics. Effectively managing these data has become essential for improving operational efficiency, reducing costs, and supporting strategic decision-making.

DataCo Global is a multinational retail and supply chain company that operates across multiple countries and markets. The company's business activities involve processing thousands of customer orders, managing diverse product categories, coordinating various shipping methods, and serving customers in different geographical regions. These operations generate large amounts of structured data that contain valuable business information but require an efficient database management system to store, organize, and analyze.

Relational database systems provide a reliable solution for handling such large-scale business data. By integrating SQL for data management and Python for data analysis and visualization, organizations can transform raw transactional data into meaningful business insights. This project implements a database system based on the DataCo Supply Chain Dataset, enabling efficient data storage, querying, and analytical reporting to support business intelligence and operational decision-making.

1.2 Problem Statement

Despite possessing extensive transactional data, many organizations face challenges in extracting valuable information for decision-making. Data is often stored in raw formats, making it difficult to perform efficient queries, identify business trends, evaluate operational performance, and generate management reports. Without a well-designed relational database, business users may experience data redundancy, inconsistent information, and inefficient analytical processes.

For the DataCo Supply Chain dataset, several critical business questions remain difficult to answer without a structured database implementation. These include identifying the most profitable products, evaluating sales performance across different markets, measuring shipping efficiency, analyzing delivery delays, and monitoring overall business profitability. Therefore, implementing a relational database integrated with SQL-based analysis and Python visualization is necessary to transform raw data into actionable business intelligence.

1.3 Project Objectives

The primary objective of this project is to implement an existing supply chain database project using PostgreSQL and Python for business data management and analysis. Specifically, the project aims to:

- Study and understand the structure of the DataCo Supply Chain dataset and also its business context.
- Design and implement a relational database schema based on the dataset using PostgreSQL.
- Import and manage structured business data while ensuring data integrity and consistency.
- Perform SQL operations, including data retrieval, aggregation, joins, and analytical queries to answer business questions.
- Connect Python to the database for data extraction, analysis, and visualization.
- Interpret analytical findings and provide business recommendations that support supply chain and sales management.

1.4 Project Scope

This project focuses on implementing an existing database-oriented analytical system using the **DataCo Supply Chain Dataset**. The scope of the project includes database design, data import, SQL querying, Python-based analysis, and business interpretation.

The database implementation covers core business entities such as customers, orders, products, categories, shipping information, markets, and geographical locations. SQL is used to perform business-oriented analyses, including sales performance, customer segmentation, product profitability, and logistics efficiency. Python is employed to generate visualizations that illustrate sales trends, product performance, delivery status, and regional business activities.

2. Project Overview

2.1 Dataset Overview

The project utilizes the **DataCo Supply Chain Dataset**, a publicly available business dataset published on Kaggle. The dataset represents transactional data generated from a global supply chain and retail business, covering customer information, product details, orders, sales performance, shipping activities, and geographical markets. It is widely used for database implementation, business intelligence, and data analytics projects because it contains realistic business scenarios and multiple interconnected entities.

The original dataset consists of over **180,000 transaction records** and **more than 50 attributes**, providing comprehensive information about different stages of the supply chain

process. Key variables include customer demographics, order details, product categories, sales revenue, profit, shipping mode, delivery status, market regions, and geographic locations. These attributes enable users to perform descriptive analysis, identify business trends, evaluate logistics performance, and support managerial decision-making.

For this project, the dataset serves as the primary data source for designing a relational database, executing SQL-based business queries, and conducting analytical visualization using Python. Since the dataset contains both operational and analytical information, it is suitable for demonstrating how database systems support business data management and reporting.

Table 2.1 Dataset Summary

Attribute	Description
Dataset Name	DataCo Supply Chain Dataset
Source	Kaggle
Data Type	Structured transactional data
Number of Records	Approximately 180,000
Number of Attributes	50+
Business Domain	Supply Chain & Retail
Main Entities	Customers, Orders, Products, Categories, Shipping, Markets
Primary Purpose	Database implementation and business analytics

2.2 Database Overview

The original dataset is provided as a single denormalized table containing all transactional information. While this format is convenient for exploratory analysis, it introduces data redundancy and increases the risk of update anomalies. To improve data integrity and support efficient querying, the dataset is transformed into a relational database following normalization principles.

The implemented database separates business information into multiple related tables representing key business entities, including customers, products, categories, orders, order details, shipping information, markets, and geographical locations. Primary keys and foreign keys are established to maintain referential integrity and accurately model the relationships between entities.

The relational database allows SQL to efficiently retrieve, aggregate, and analyze data across multiple tables using JOIN operations. This structure also improves scalability, minimizes duplicate information, and provides a foundation for business intelligence reporting.

2.3 Entity Relationship Diagram (ERD) / Database Schema

To represent the business process, the database is organized into several core entities. The central entity is **Orders**, which connects customers, products, and shipping information through relational keys.

The proposed database schema consists of the following tables:

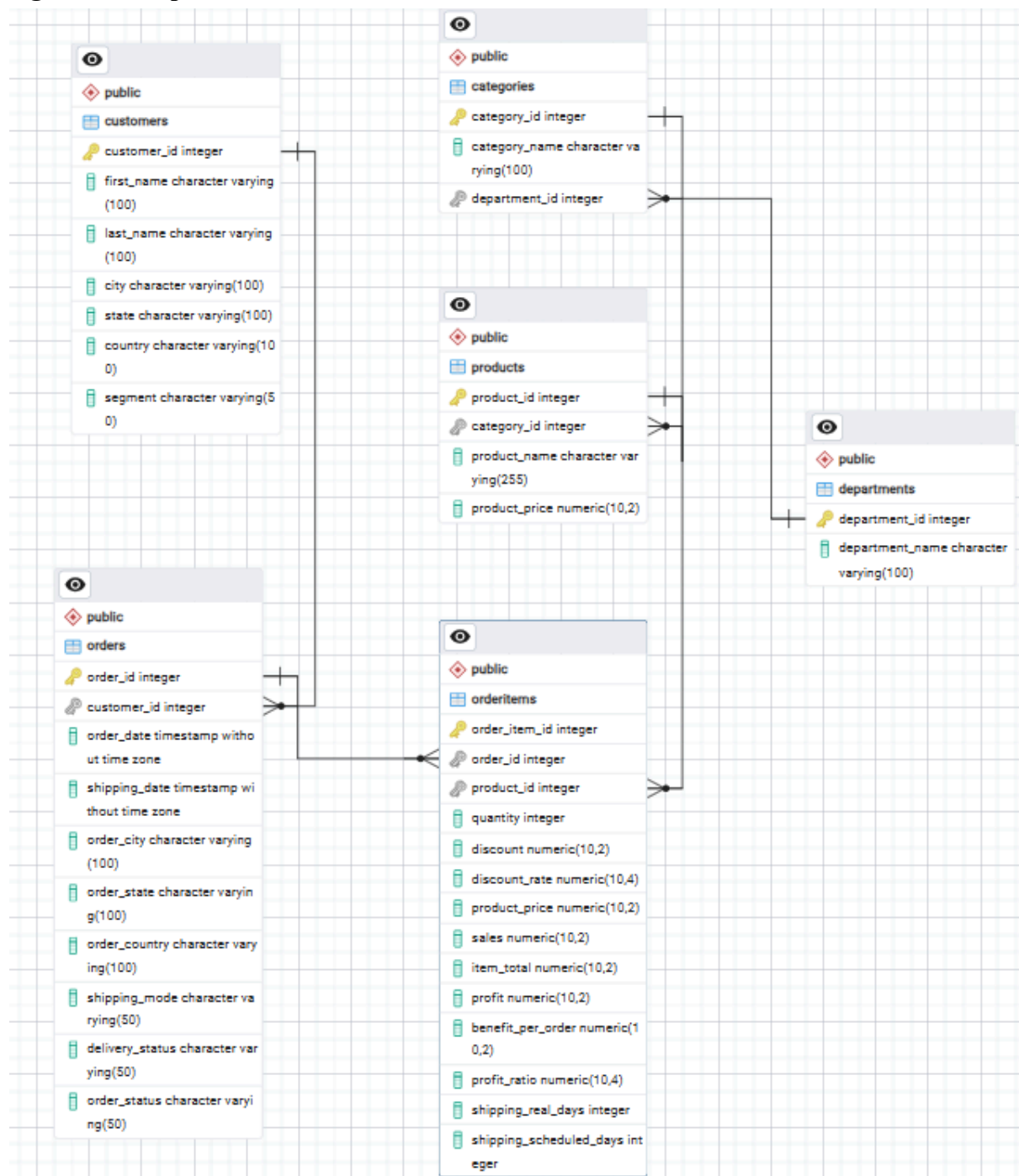
Table	Description
Customer	Stores customer profile and location information
Orders	Stores order transactions and order dates
Order_Details	Records purchased products, quantities, sales, discounts, and profits
Product	Stores product information and product prices

Category	Classifies products into business categories
Shipping	Stores shipping mode, delivery status, and delivery performance
Market	Stores market region information
Geography	Stores country, state, city, latitude, and longitude information

The relationships between these entities can be summarized as follows:

- One customer can place multiple orders.
- One order can contain multiple products.
- Each product belongs to one category.
- Each order uses one shipping method.
- Orders are associated with a specific market and geographical location.

Figure 2.1 Proposed Database Schema



2.4 Tools and Technologies

Several software tools and programming technologies are employed throughout this project to support database implementation, data analysis, and visualization.

Table 2.2 Tools and Technologies

Tool / Technology	Purpose
PostgreSQL	Relational database management system used to implement the database schema and store business data
pgAdmin 4	Graphical interface for database administration and SQL execution
SQL	Data definition, manipulation, and analytical querying
Python	Data extraction, preprocessing, analysis, and visualization
Pandas	Data manipulation and database connectivity
Matplotlib	Data visualization
Microsoft Excel	Initial inspection and verification of dataset quality
GitHub	Version control and project repository management

3. SQL-Based Data Analysis

This chapter presents the SQL-based analyses conducted on the implemented PostgreSQL database. The queries are designed to answer key business questions related to overall business performance, product sales, customer behavior, logistics efficiency, and sales trends. The findings provide managers with valuable insights for strategic decision-making.

3.1 Business Question 1

How is the overall business performing in terms of sales, profit, and average order value?

Objective

To evaluate the company's overall financial performance using three key performance indicators (KPIs): total sales, total profit, and average order value.

SQL Query:

```
SELECT ROUND(SUM(sales), 2) AS total_sales
FROM OrderItems;

SELECT ROUND(SUM(profit), 2) AS total_profit
FROM OrderItems;

SELECT
    ROUND(AVG(item_total), 2) AS average_order_value
FROM OrderItems;
```

SQL Output:

	total_sales numeric		total_profit numeric
1	36784734.31	1	3966902.97

	average_order_value numeric
1	183.11

Business Insight:

The company's overall performance indicates a healthy business operation with **total sales of approximately \$36.78 million** and **total profit of nearly \$3.97 million**, resulting in an estimated **profit margin of around 10.8%**. Meanwhile, the **average order value of \$183.11** suggests that customers typically purchase multiple products or relatively high-value items in each transaction.

These KPIs provide management with an overall view of business performance and serve as baseline indicators for evaluating future growth. To improve profitability, management should focus not only on increasing sales volume but also on optimizing pricing strategies, reducing unnecessary discounts, and improving operational efficiency to increase profit margins.

3.2 Business Question 2

Which products and product categories contribute the most to sales?

Objective

To identify high-performing and low-performing products, as well as evaluate revenue contribution by category and department.

- **Top 10 Best-selling Products**

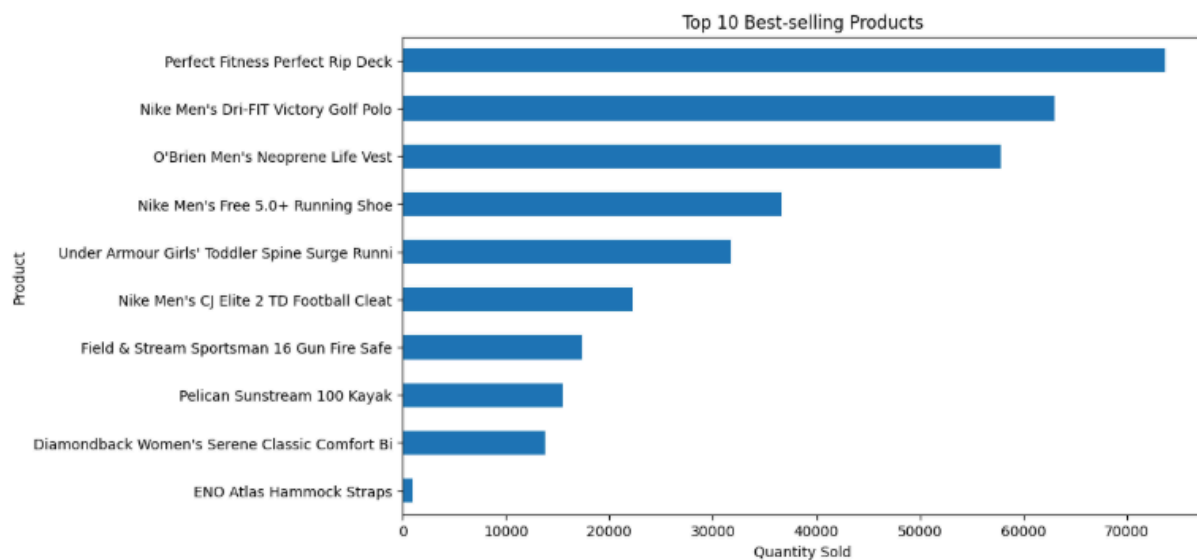
SQL Query:

```
SELECT
    p.product_name,
    SUM(oi.quantity) AS quantity_sold
FROM OrderItems oi
JOIN Products p
    ON oi.product_id = p.product_id
GROUP BY p.product_name
ORDER BY quantity_sold DESC
LIMIT 10;
```

SQL Output:

	product_name character varying (255)	quantity_sold bigint
1	Perfect Fitness Perfect Rip Deck	73698
2	Nike Men's Dri-FIT Victory Golf Polo	62956
3	O'Brien Men's Neoprene Life Vest	57803
4	Nike Men's Free 5.0+ Running Shoe	36680
5	Under Armour Girls' Toddler Spine Surge Runni	31735
6	Nike Men's CJ Elite 2 TD Football Cleat	22246
7	Field & Stream Sportsman 16 Gun Fire Safe	17325
8	Pelican Sunstream 100 Kayak	15500
9	Diamondback Women's Serene Classic Comfor...	13729
10	ENO Atlas Hammock Straps	998

Visualization:



- Bottom 10 Least-selling Products

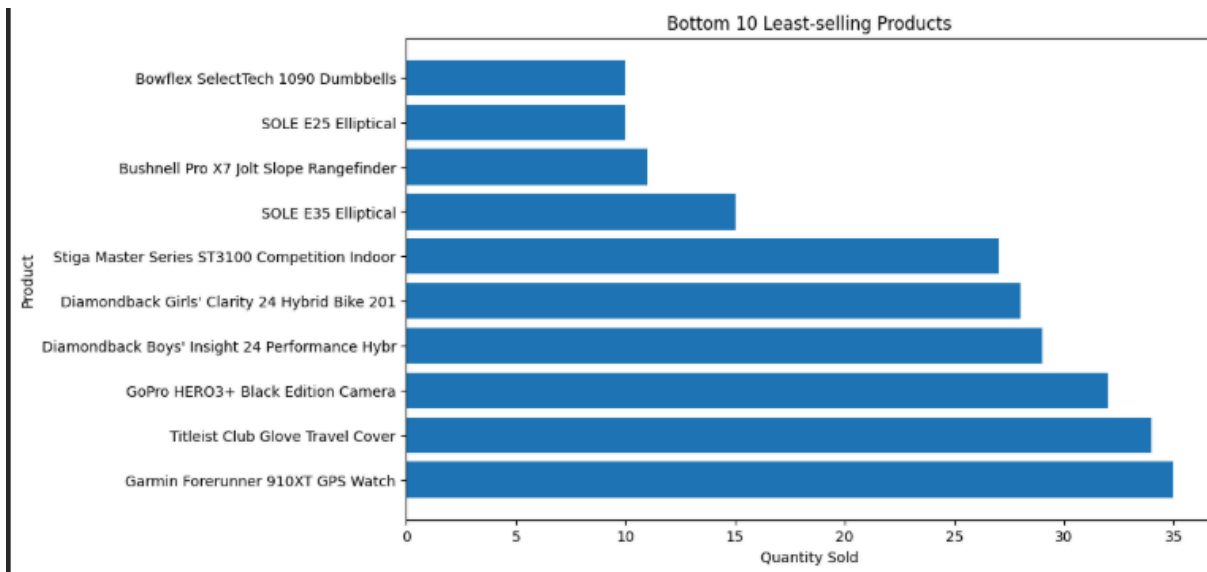
SQL Query:

```
SELECT
    p.product_name,
    SUM(oi.quantity) AS quantity_sold
FROM OrderItems oi
JOIN Products p
    ON oi.product_id = p.product_id
GROUP BY p.product_name
ORDER BY quantity_sold ASC
LIMIT 10;
```

SQL Output:

	product_name character varying (255)	quantity_sold bigint
1	Bowflex SelectTech 1090 Dumbbells	10
2	SOLE E25 Elliptical	10
3	Bushnell Pro X7 Jolt Slope Rangefinder	11
4	SOLE E35 Elliptical	15
5	Stiga Master Series ST3100 Competition Indoor	27
6	Diamondback Girls' Clarity 24 Hybrid Bike 201	28
7	Diamondback Boys' Insight 24 Performance H...	29
8	GoPro HERO3+ Black Edition Camera	32
9	Titleist Club Glove Travel Cover	34
10	Garmin Forerunner 910XT GPS Watch	35

Visualization:



● Revenue by Product Category

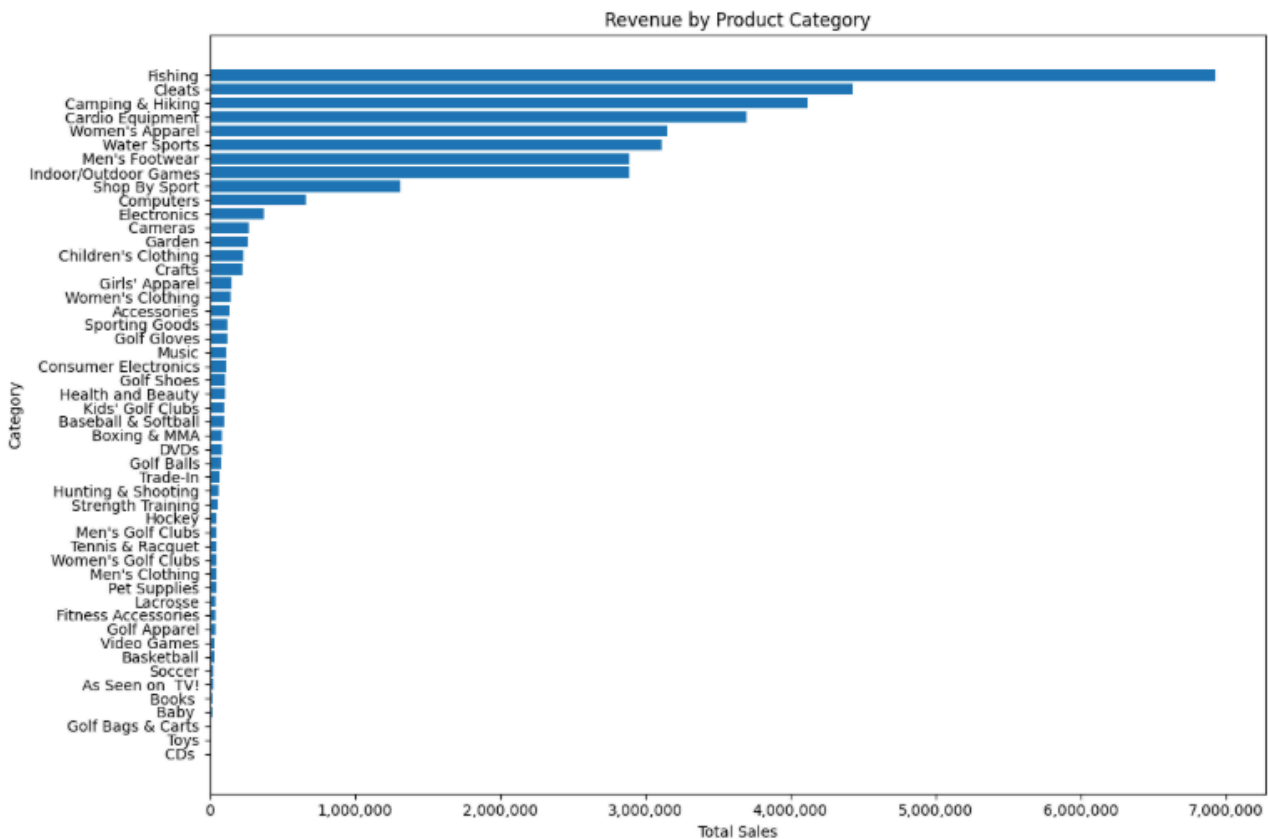
SQL Query:

```
SELECT
    ca.category_name,
    ROUND(SUM(oi.sales), 2) AS total_sales
FROM OrderItems oi
JOIN Products p
    ON oi.product_id = p.product_id
JOIN Categories ca
    ON p.category_id = ca.category_id
GROUP BY ca.category_name
ORDER BY total_sales DESC;
```

SQL Output:

	category_name character varying (100) 🔒	total_sales numeric 🔒
1	Fishing	6929653.50
2	Cleats	4431942.66
3	Camping & Hiking	4118425.42
4	Cardio Equipment	3694843.20
5	Women's Apparel	3147800.00
6	Water Sports	3113844.60
7	Men's Footwear	2891757.54
8	Indoor/Outdoor Games	2888993.94
9	Shop By Sport	1309522.02
10	Computers	663000.00
11	Electronics	371034.64
12	Cameras	267607.68

Visualization:



Revenue by Department

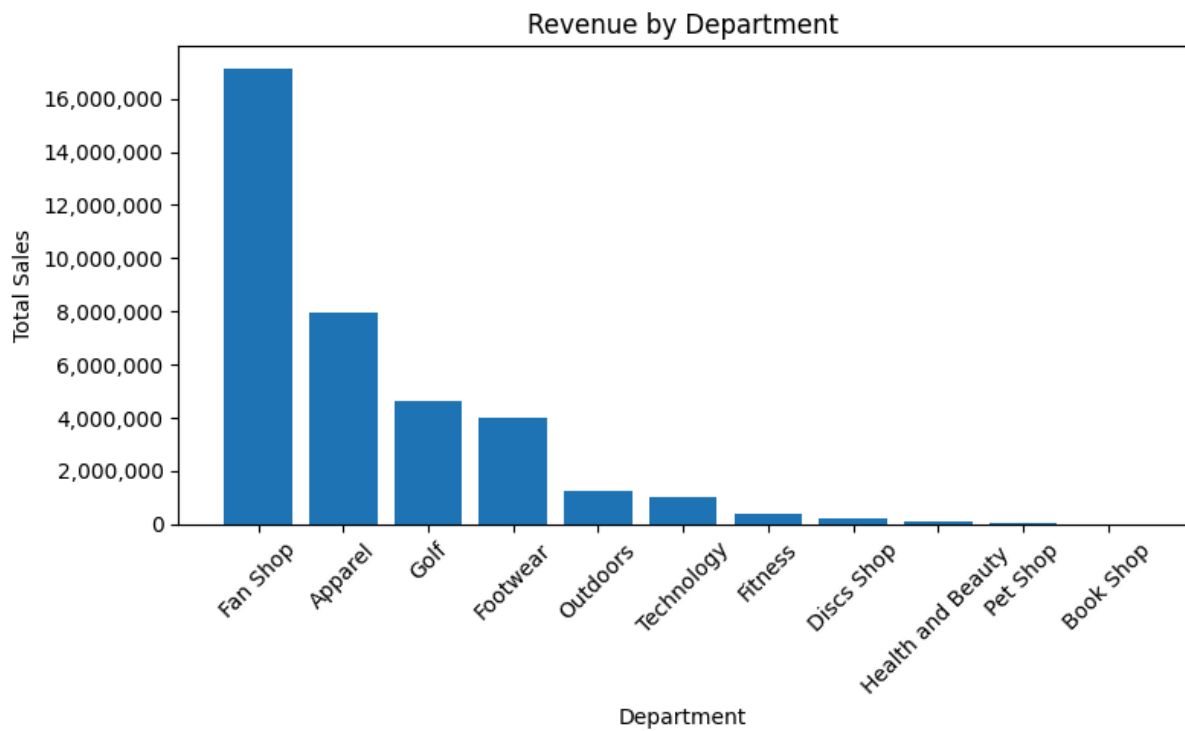
SQL Query:

```
SELECT
    d.department_name,
    ROUND(SUM(oi.sales),2) AS revenue
FROM Departments d
JOIN Categories c
ON d.department_id = c.department_id
JOIN Products p
ON c.category_id = p.category_id
JOIN OrderItems oi
ON p.product_id = oi.product_id
GROUP BY d.department_name
ORDER BY revenue DESC;
```

SQL Output:

	department_name character varying (100) 🔒	revenue numeric 🔒
1	Fan Shop	17113870.54
2	Apparel	7976255.08
3	Golf	4609028.22
4	Footwear	4006498.77
5	Outdoors	1253351.44
6	Technology	1039598.96
7	Fitness	397050.89
8	Discs Shop	228887.73
9	Health and Beauty	106080.48
10	Pet Shop	41524.80
11	Book Shop	12587.40

Visualization:



Business Insight

The analysis reveals a highly uneven distribution of product performance across the catalog. Products such as **Perfect Fitness Perfect Rip Deck** and **Nike Men's Dri-FIT Victory Golf Polo** dominate total sales volume, indicating strong customer demand and consistent purchasing behavior. These products should receive priority in inventory planning, promotional campaigns, and replenishment strategies to avoid stock shortages.

At the category level, **Fishing, Cleats, and Camping & Hiking** generate the highest revenue, suggesting that outdoor and sporting products represent the company's core revenue drivers. Conversely, categories such as **Cameras, Electronics, and Computers** contribute relatively little to overall sales and should be reviewed for pricing effectiveness, product assortment, and marketing support.

Department-level analysis further confirms that the **Fan Shop** department contributes substantially more revenue than other departments, indicating a high concentration of business performance within a limited number of product groups. Management should continue investing in high-performing departments while evaluating whether low-performing departments require restructuring, targeted promotions, or product rationalization to improve overall profitability.

3.3 Business Question 3

What are the purchasing patterns of different customer segments and geographical regions?

Objective

To understand customer purchasing behavior across different market segments and identify the regions generating the highest sales.

- **Sales by Customer Segment**

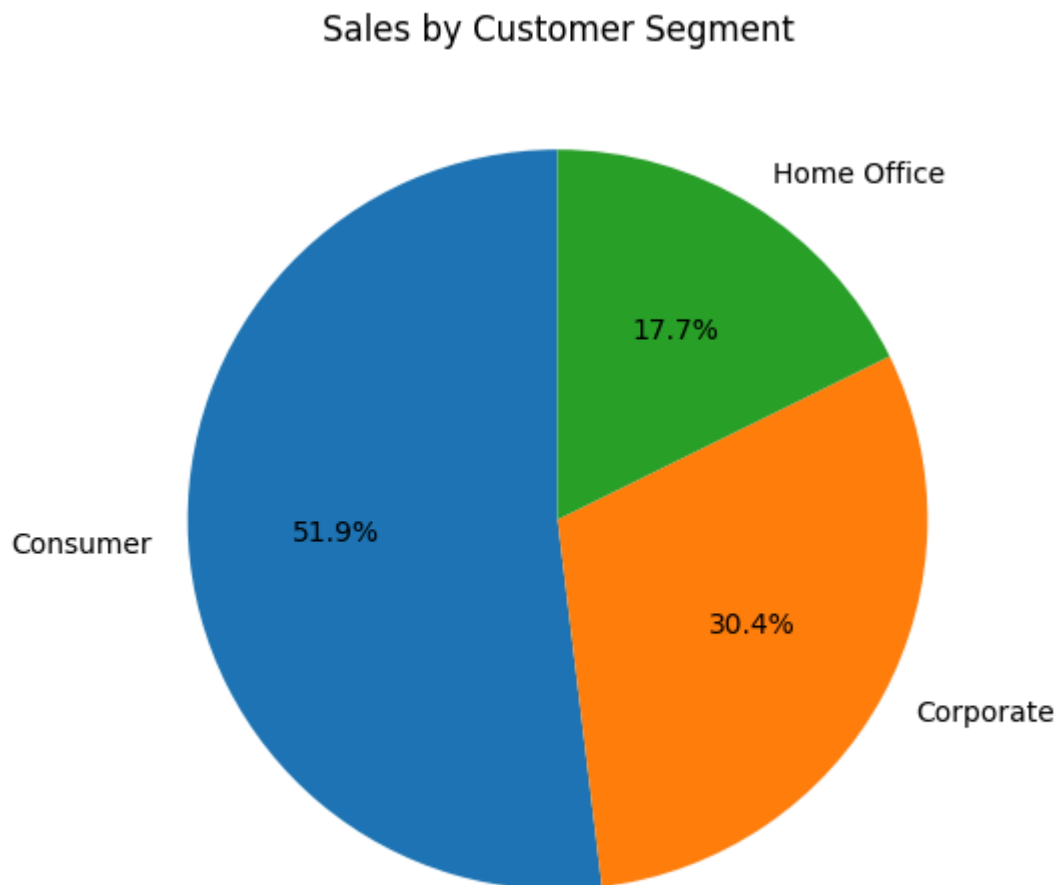
SQL Query:

```
SELECT
    c.segment,
    ROUND(SUM(oi.sales),2) AS total_sales
FROM Customers c
JOIN Orders o
ON c.customer_id = o.customer_id
JOIN OrderItems oi
ON o.order_id = oi.order_id
GROUP BY c.segment
ORDER BY total_sales DESC;
```

SQL Output:

	segment character varying (50) 🔒	total_sales numeric 🔒
1	Consumer	19095789.79
2	Corporate	11168406.63
3	Home Office	6520537.89

Visualization:



- Top 10 States by Sales

SQL Query:

```

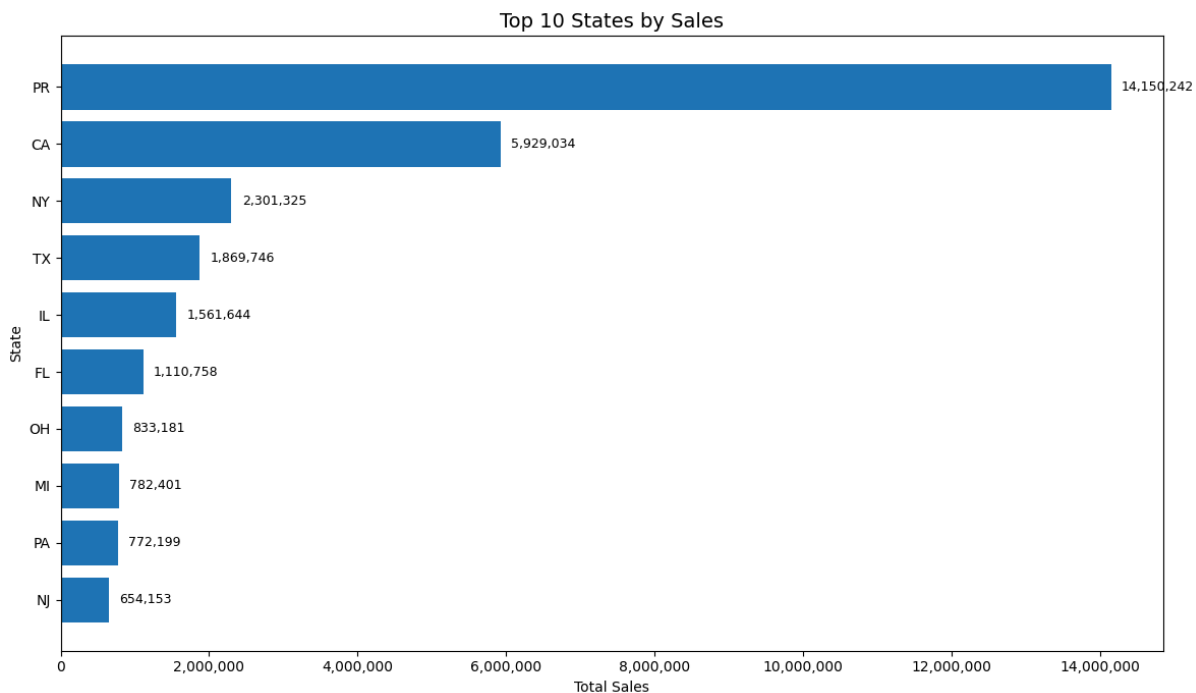
SELECT
    c.state,
    ROUND(SUM(oi.sales), 2) AS total_sales
FROM Customers c
JOIN Orders o
    ON c.customer_id = o.customer_id
JOIN OrderItems oi
    ON o.order_id = oi.order_id
GROUP BY c.state
ORDER BY total_sales DESC
LIMIT 10;

```

SQL Output:

	state character varying (100) 🔒	total_sales numeric 🔒
1	PR	14150241.50
2	CA	5929033.90
3	NY	2301325.11
4	TX	1869746.06
5	IL	1561644.50
6	FL	1110758.39
7	OH	833180.84
8	MI	782401.30
9	PA	772199.00
10	NJ	654152.53

Visualization:



Business Insight

Customer purchasing behavior varies significantly across market segments. The **Consumer** segment contributes the largest proportion of total sales, considerably outperforming both the **Corporate** and **Home Office** segments. This indicates that individual consumers remain the company's primary source of revenue and should remain the primary focus of marketing investments, customer engagement programs, and loyalty initiatives.

Geographical analysis shows that sales are heavily concentrated in several states, particularly **Puerto Rico (PR)** and **California (CA)**, followed by New York and Texas. These high-performing markets represent strategic regions where additional inventory allocation, localized marketing campaigns, and logistics resources may generate even greater returns.

Meanwhile, states with relatively lower sales may require market-specific promotional strategies or further investigation to determine whether low demand is caused by limited customer penetration, supply chain constraints, or regional competition.

3.4 Business Question 4

How efficient is the company's logistics and delivery performance?

Objective

To evaluate shipping performance by analyzing shipping methods, delivery status, and delivery punctuality.

- Shipping Mode Distribution

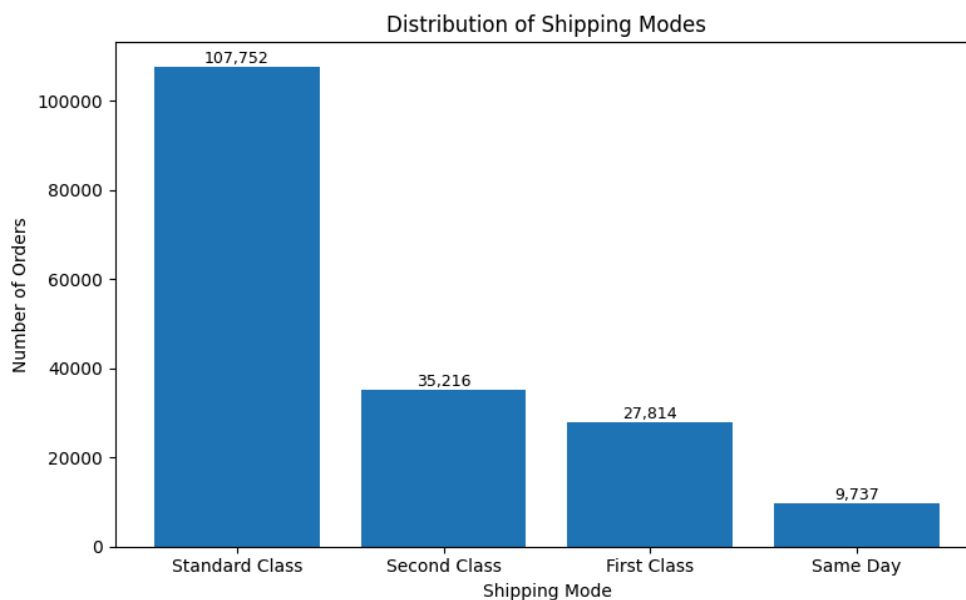
SQL Query:

```
SELECT
    o.shipping_mode,
    COUNT(oi.order_item_id) AS total_items
FROM OrderItems oi
JOIN Orders o
    ON oi.order_id = o.order_id
GROUP BY o.shipping_mode
ORDER BY total_items DESC;
```

SQL Output:

	shipping_mode character varying (50)	total_items bigint
1	Standard Class	107752
2	Second Class	35216
3	First Class	27814
4	Same Day	9737

Visualization:



- Delivery Status Distribution

SQL Query:

```

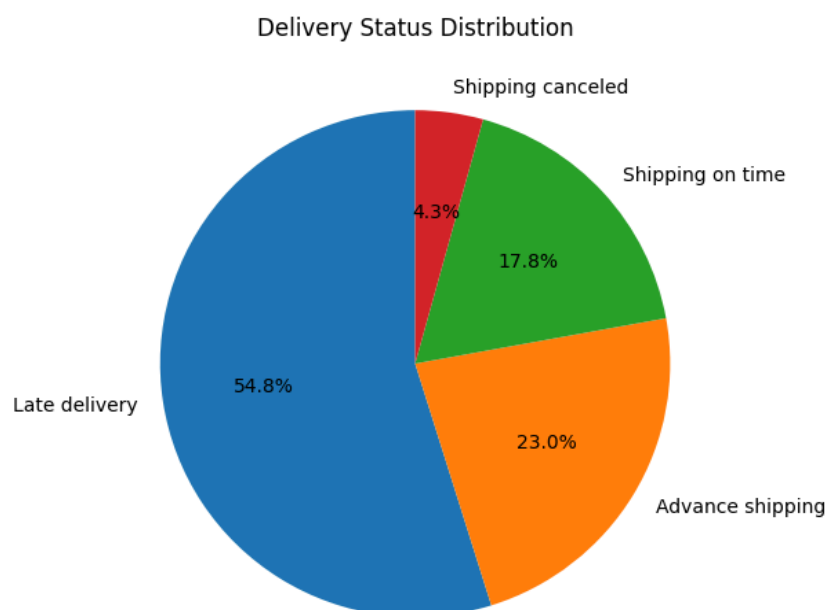
SELECT
    o.delivery_status,
    COUNT(oi.order_item_id) AS total_items
FROM OrderItems oi
JOIN Orders o
    ON oi.order_id = o.order_id
GROUP BY o.delivery_status
ORDER BY total_items DESC;

```

SQL Output:

	delivery_status character varying (50) 🔒	total_items bigint 🔒
1	Late delivery	98977
2	Advance shipping	41592
3	Shipping on time	32196
4	Shipping canceled	7754

Visualization:



- On-time vs Delayed Deliveries

SQL Query:

```

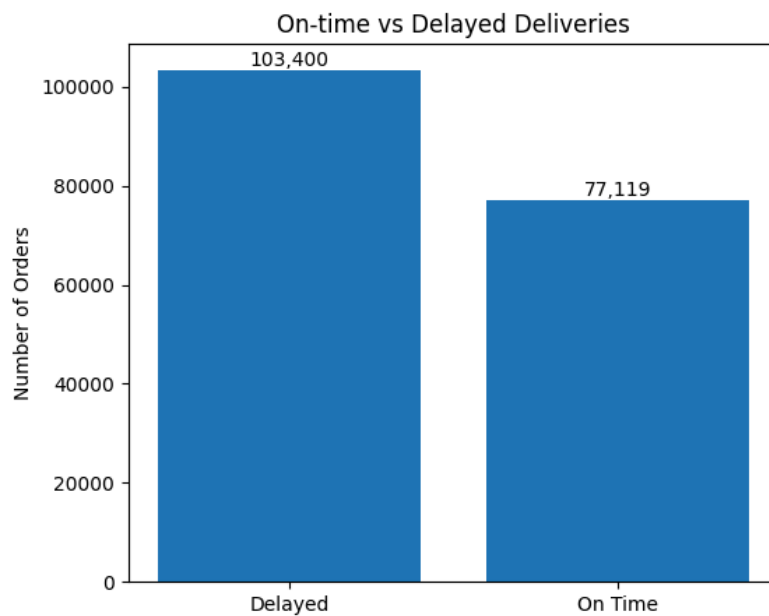
SELECT
    CASE
        WHEN oi.shipping_real_days <= oi.shipping_scheduled_days
            THEN 'On Time'
        ELSE 'Delayed'
    END AS delivery_performance,
    COUNT(*) AS total_orders
FROM OrderItems oi
GROUP BY delivery_performance
ORDER BY total_orders DESC;

```

SQL Output:

	delivery_performance 	total_orders 
1	Delayed	103400
2	On Time	77119

Visualization:



Business Insight

The logistics analysis indicates several operational challenges that directly affect customer service quality. Although **Standard Class** is the most frequently used shipping method, the dataset also shows that **late deliveries represent the largest delivery status**, highlighting inefficiencies within the company's fulfillment and transportation processes.

Furthermore, the comparison between scheduled and actual shipping days reveals that **delayed deliveries occur more frequently than on-time deliveries**, suggesting that current logistics operations may not consistently meet customer expectations. Persistent delivery delays can negatively influence customer satisfaction, increase cancellation risk, and reduce repeat purchase behavior.

To improve operational performance, management should investigate the root causes of delivery delays, evaluate carrier performance, optimize warehouse operations, and review shipping schedules. Establishing logistics performance KPIs such as on-time delivery rate and average delivery delay would support continuous performance monitoring and service improvement.

3.5 Business Question 5

How do sales and profit change over time?

Objective

To analyze monthly trends in sales and profitability to support forecasting and business planning.

SQL Query:


```

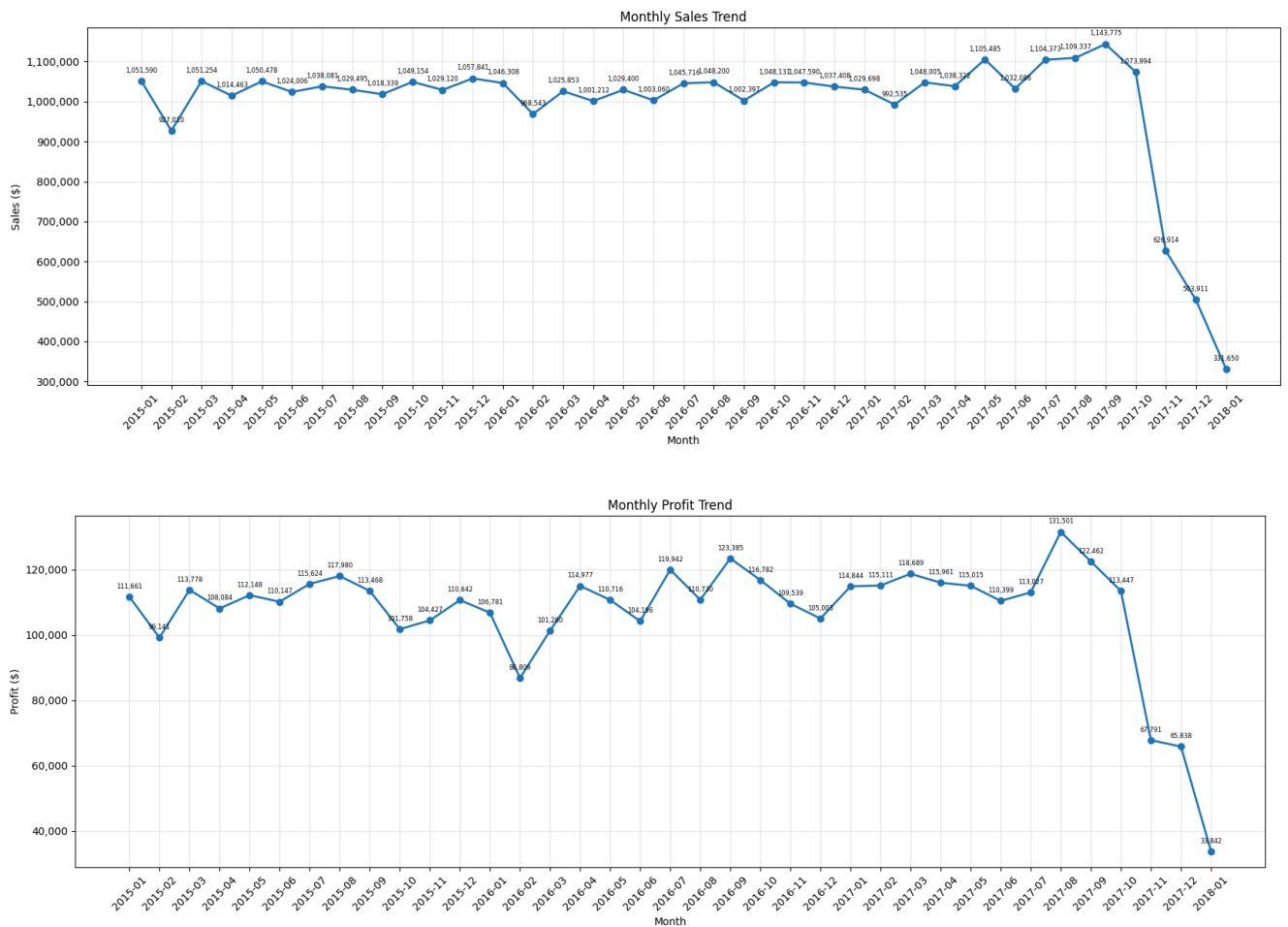
SELECT
    DATE_TRUNC('month', o.order_date) AS month,
    ROUND(SUM(oi.sales), 2) AS total_sales,
    ROUND(SUM(oi.profit), 2) AS total_profit
FROM Orders o
JOIN OrderItems oi ON o.order_id = oi.order_id
GROUP BY DATE_TRUNC('month', o.order_date)
ORDER BY month;
--Download dataset to convert into python
SELECT
    oi.order_item_id,o.order_id,o.order_date,o.shipping_date, c.customer_id,
    c.first_name,c.last_name,c.segment,c.city,c.state,c.country,
    d.department_name,ca.category_name,p.product_name,p.product_price,
    oi.quantity, oi.sales,oi.item_total,oi.profit,oi.discount,oi.discount_rate,
    oi.shipping_real_days,oi.shipping_scheduled_days,o.shipping_mode,
    o.delivery_status,o.order_status
FROM OrderItems oi
JOIN Orders o ON oi.order_id = o.order_id
JOIN Customers c ON o.customer_id = c.customer_id
JOIN Products p ON oi.product_id = p.product_id
JOIN Categories ca ON p.category_id = ca.category_id
JOIN Departments d ON ca.department_id = d.department_id;

```

SQL Output:

	month timestamp without time zone	total_sales numeric	total_profit numeric
1	2015-01-01 00:00:00	1051590.06	111660.74
2	2015-02-01 00:00:00	927009.88	99140.66
3	2015-03-01 00:00:00	1051253.67	113778.21
4	2015-04-01 00:00:00	1014463.26	108083.68
5	2015-05-01 00:00:00	1050478.42	112147.90
6	2015-06-01 00:00:00	1024006.15	110147.16
7	2015-07-01 00:00:00	1038081.17	115624.06
8	2015-08-01 00:00:00	1029494.67	117979.77
9	2015-09-01 00:00:00	1018338.58	113467.94
10	2015-10-01 00:00:00	1049154.25	101757.87
11	2015-11-01 00:00:00	1029120.22	104427.36
12	2015-12-01 00:00:00	1057840.86	110641.55

Visualization:



Business Insight

Monthly sales and profit remain relatively stable throughout the observed period, with moderate fluctuations rather than dramatic seasonal changes. This suggests that customer demand is relatively consistent over time, providing the company with a stable revenue base for operational planning.

Although sales fluctuate slightly between months, profitability generally follows a similar trend, indicating that profit performance is closely associated with sales volume. However, months with lower profit despite relatively stable sales may indicate increased discounting, higher operating costs, or changes in product mix.

Continuous monitoring of monthly sales and profit trends enables management to identify seasonal demand patterns, forecast future performance more accurately, optimize inventory planning, and schedule promotional activities during periods of slower business activity. Long-term trend analysis can also support budgeting and strategic planning decisions.

4. Business Recommendations

Based on the SQL and Python analyses, four key recommendations are proposed.

4.1. Improve profitability through product and pricing management

With approximately \$36.78 million in sales and \$3.97 million in profit, DataCo's estimated profit margin is approximately 10.8%. Management should therefore focus on profitable growth by monitoring product-level margins, controlling excessive discounts, and prioritizing products that combine strong demand with attractive profitability.

4.2. Optimize inventory and product portfolio allocation

Sales are concentrated in a limited number of products and categories, particularly Fishing, Cleats and Camping & Hiking. High-performing products should receive priority in inventory planning and replenishment, while consistently underperforming categories should be reviewed for pricing, assortment and demand before additional resources are allocated.

4.3. Improve delivery reliability

Late delivery represents a major operational concern in the dataset. Management should establish logistics KPIs such as On-Time Delivery Rate, Late Delivery Rate and Average Delivery Delay, and analyze performance by shipping mode, region, product and customer segment. This will help identify operational bottlenecks and improve delivery reliability.

4.4. Adopt segment- and region-specific customer strategies

The Consumer segment is the company's largest revenue contributor, while sales are concentrated in several major geographical markets. DataCo should prioritize customer retention in the Consumer segment, develop targeted approaches for Corporate and Home Office customers, and allocate inventory and logistics resources according to regional demand and operational performance.

5. Conclusion

This project successfully developed a normalized PostgreSQL database and Python analytical pipeline based on the DataCo Global supply chain dataset (~180,000 records), optimizing data storage and enabling key business insights. The analysis revealed a total revenue of \$36.78M and a profit of \$3.97M, driven primarily by the Consumer segment, top geographical markets in Puerto Rico and California, and leading categories such as Fishing, Cleats, and Fan Shop. Crucially, the system exposed critical logistics bottlenecks, particularly a high frequency of delivery delays despite the prevalence of Standard Class shipping. By addressing data redundancy and establishing robust business metrics, this study provides actionable strategic recommendations for inventory optimization, product portfolio rationalization, and supply chain fulfillment enhancements, while laying a solid foundation for future integration with predictive machine learning models and real-time BI dashboards.